Customer segmentation with clustering

(Shutterstock, 2024)

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C101 – Week 6 Mini-Project

Authored by: Jason Kurth

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Context and Problem statement

Unsupervised machine learning methods are effective to analyse customer clustering. Using this company's data, customer cohorts were created with characteristics around shopping habits.

With clear definitions of the types of customers the business attracts, the company can better target its promotions, implement efficient retention strategies and improve customer experience, in turn increasing profitability.

Methodology

The data included almost one million rows and 19 features. After removing duplicates, through feature engineering, data was organised by customer, reducing to information to 68,000 rows, with five columns of relevant customer data: frequency, recency, customer lifetime value (CLV), average unit cost (AUC) and age.

Descriptive statistics and visualisations were created to understand the distribution and outlier percentages.

Due to computational constraints, clustering was performed on a random sample of 10,000 customers.

Three methodologies were used to determine the optimal number of clusters for this data:

Table 1. Optimal # of clusters determination method summary

Determination Method	Description	Evaluation / Decision Criteria
1. K-means – Elbow Method	Uses within-cluster sum of squares (inertia) to measure how compact clusters are as the number of clusters k increases.	Location of the “elbow” point where additional clusters result in only marginal reductions in inertia
2. K-means – Silhouette Score	Measures how well each data point fits within its assigned cluster compared to other clusters	Silhouette score. Higher values indicate better defined clusters.
3. Hierarchical Clustering - Dendrogram	Builds a hierarchy of clusters using pairwise distances and linkage criteria, visualised as a dendrogram	Long linkage distance suggests natural cut points that indicate a suitable number of clusters.

Finally, data was scaled, and PCA and t-SNE were run. The PCA’s explained variance and visualisation was used to determine the proportion of the data explained by each component and evaluate cluster separation. T-SNE was used to see if local neighbourhoods were preserved within clusters.

Results

EDA findings

Boxplots were created to help understand initial data patterns.

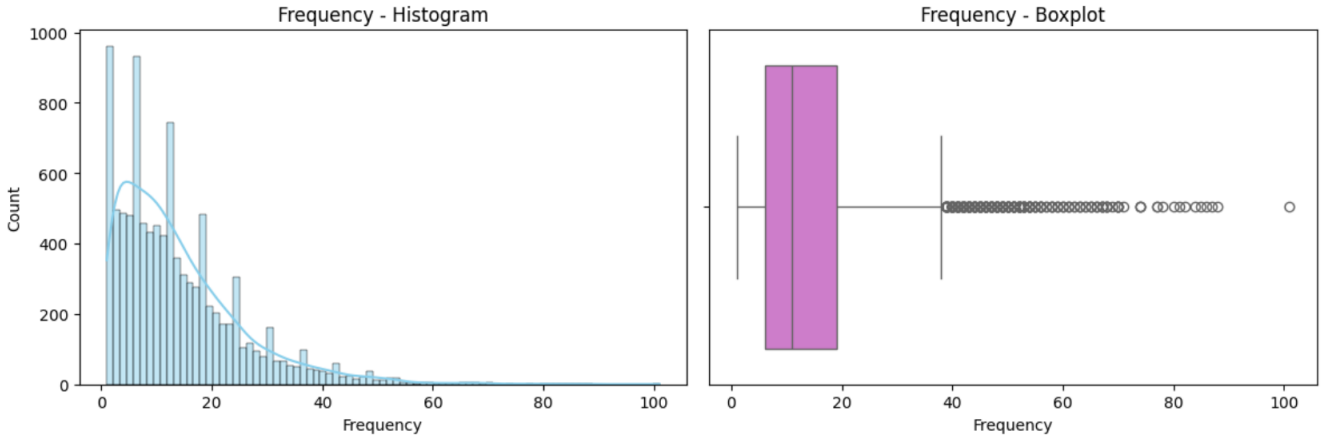


Figure 1. Frequency - Most customers make 1-40 purchases; few exceeded 120+, producing many outliers (4.16%)

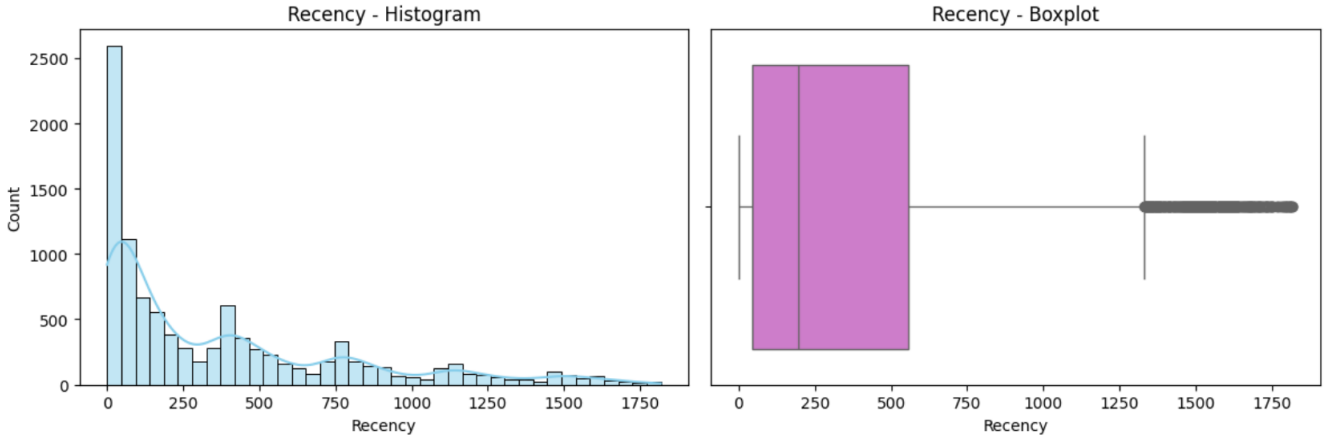


Figure 2. Recency – Spikes from seasonality effects. Multiple outliers (4.64%)

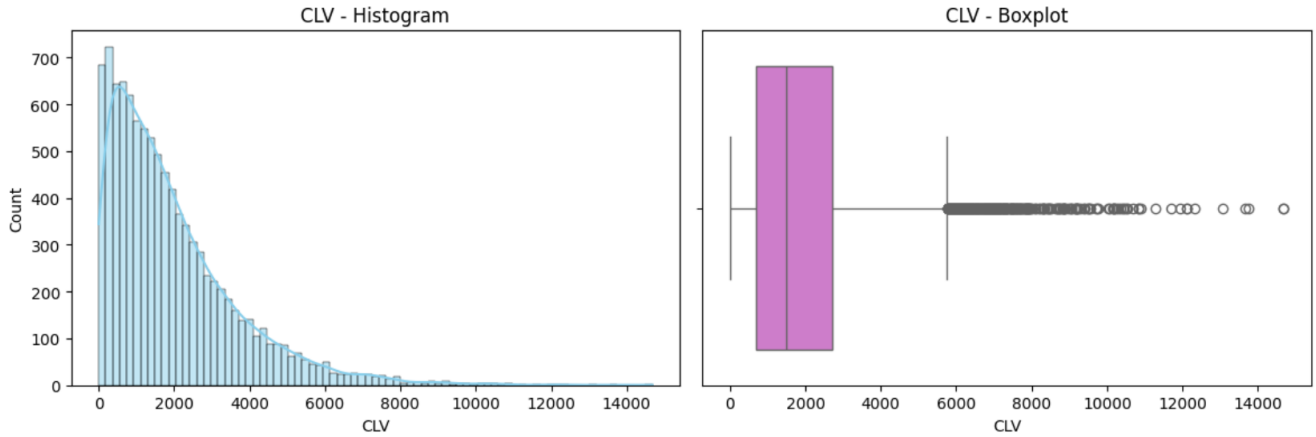


Figure 3. CLV - Strongly right-skewed, small number of extremely high-value customers; numerous outliers (3.83%)

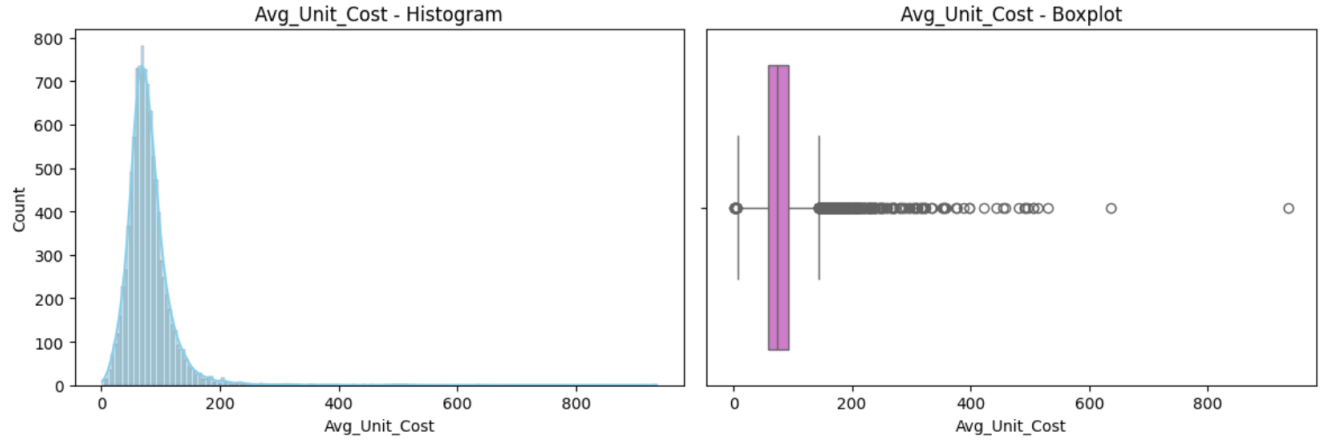


Figure 4. AUC - Approximately bell-shaped distribution with a long right tail; many high-price outliers (4.22%)

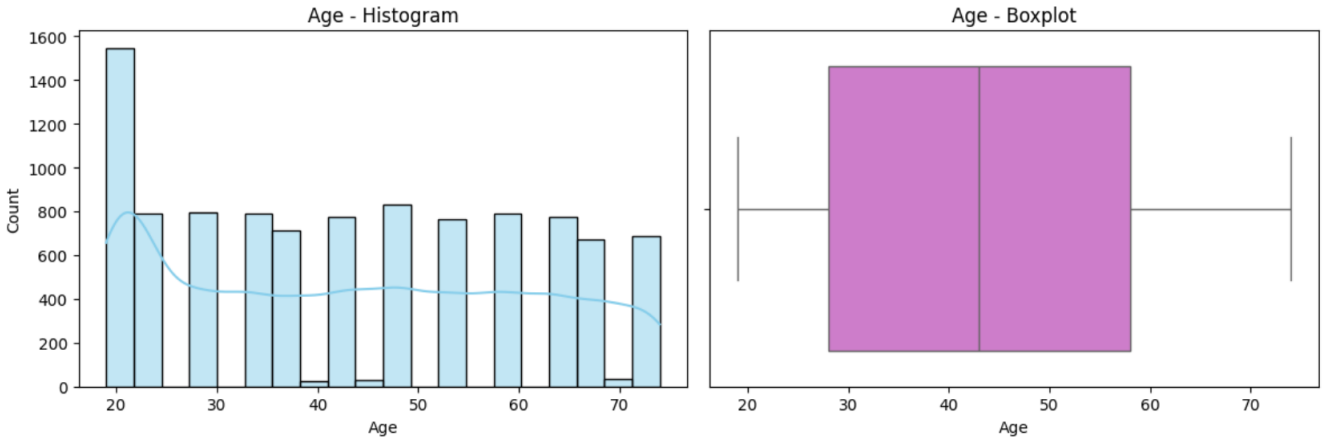


Figure 5. Age - Customer ages from a peak at 18 to over 70 years (no outliers)

Ideal cluster number findings

Elbow method

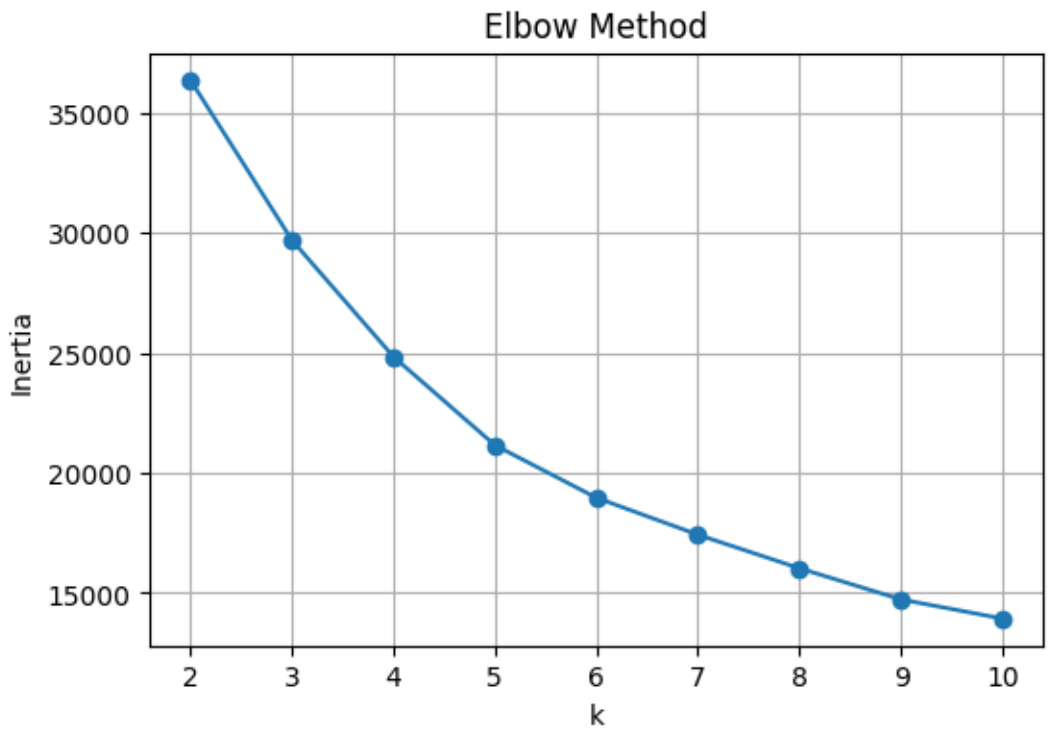


Figure 6. Elbow Method output for 2-10 clusters

Steepest slope between 2-3 clusters, and progressively becomes less steep. Due a slow rate of change in the slope angle, between 3-6 could be an appropriate number of clusters; 3 has a significantly lower inertia than 2, and from 6 onwards, inertia does not change much anymore. Silhouette Score will use this range to help us narrow down our ideal k.

Silhouette Score

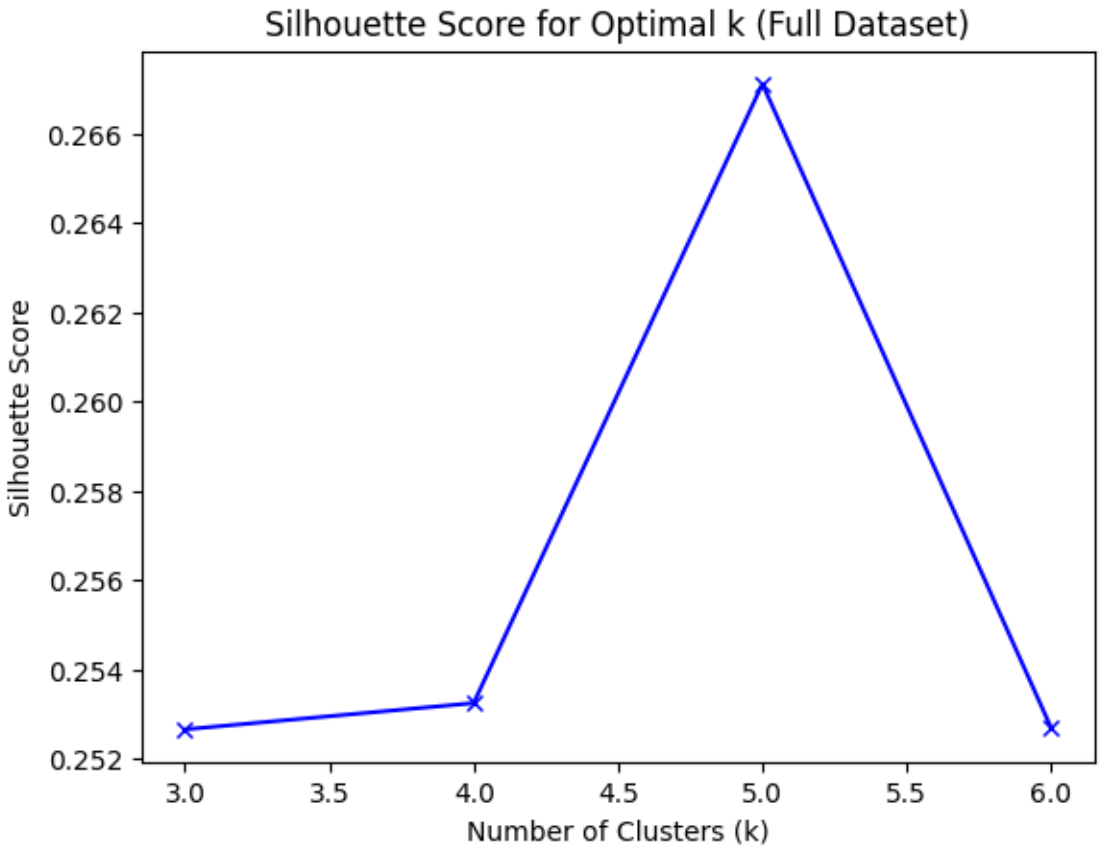
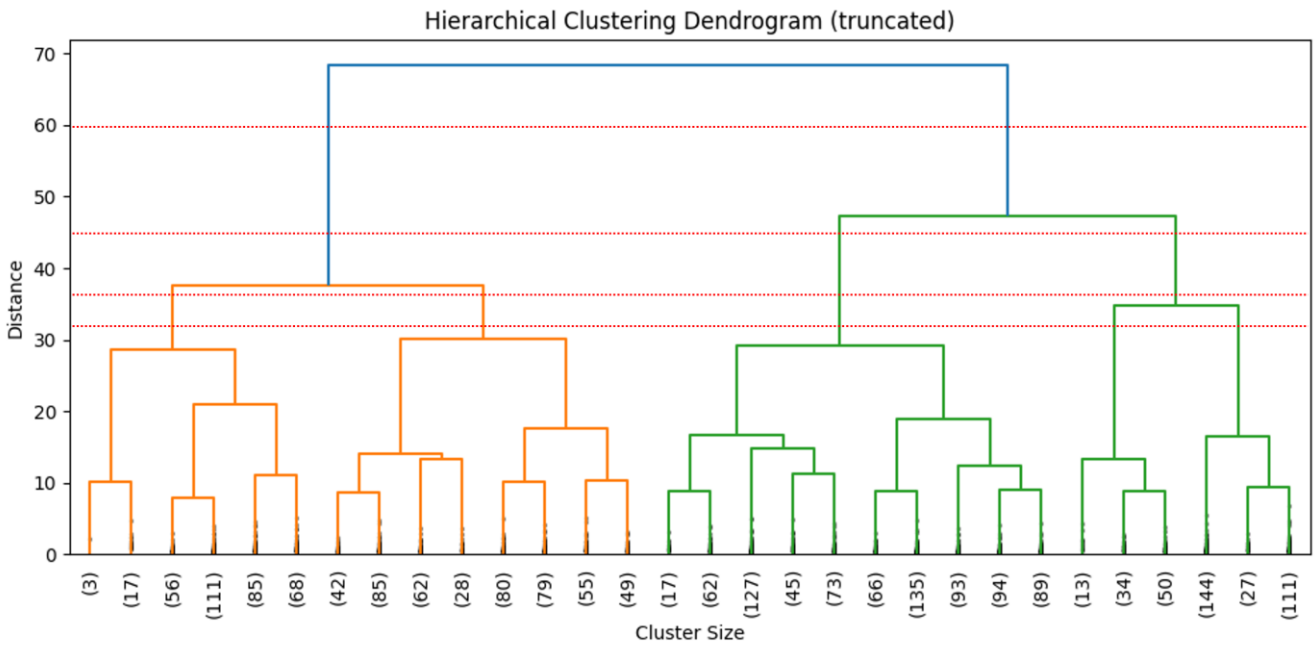


Figure 7. Silhouette Score method output for 3-6 clusters

The silhouette score is best at 5, however, the scores are all quite low around the .25-.27 mark. This is a low cluster score, and therefore means weak evidence for choosing 5, even though it is definitely the highest.

Hierarchical Clustering

Figure 8. Dendrogram by Hierarchical Clustering



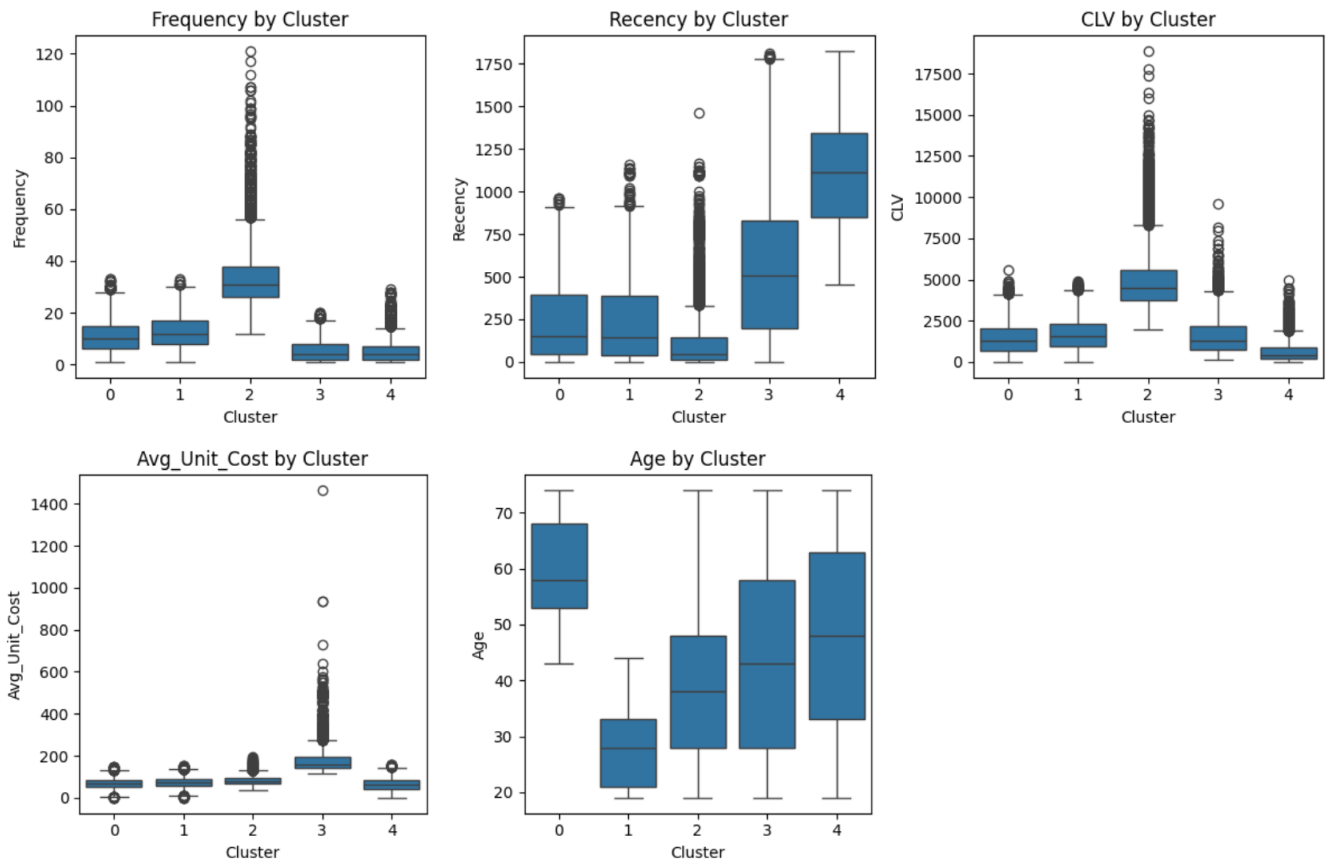
2 long linkages appear at 60, indicating 2 are "enough" clusters; however, based on past analysis in the elbow method, we concluded 2 is too few. There are 3 long linkages at 45, indicating a strong option. At 37 and 32, there are 4 & 5 clusters respectively; the orange lines are quite short, showing weak grouping, but all 2/3 green linkages are long, suggesting strongly dissimilar clusters. From the dendrogram and previous analysis, 3-5 clusters is our best estimate for ideal k.

Method	**k range evaluated**	**Key signal**	***Suggested k***
Elbow Method	2–10	Inertia reduction slows after k≈6	*3-6*
Silhouette Score	3–6	Highest score at k=5	*5*
Hierarchical Clustering	full data (sampled)	Length of linkages	*3-5*
Final decision		Summary of results across methods	*5*

Based on all three clustering methods, 3-5 is an acceptable range of choices. However, based on Silhouette Score, 5 was used as optimal number of k.

Cluster Profiles

I ran the k-means clustering at k=5 and created box plot charts for each variable, allowing me to take insights and create customer profiles for each cohort.



	Frequency		Recency		CLV		Avg_Unit_Cost		Age	
	median	std	median	std	median	std	median	std	median	std
Cluster										
0	10.0	6.151176	150.0	212.148661	1273.34	910.715068	69.277500	24.068501	58.0	8.910759
1	12.0	6.082370	142.0	225.119903	1581.53	905.251998	72.073252	23.651704	28.0	7.968075
2	31.0	11.371713	44.0	177.296618	4481.59	1694.967619	80.094545	19.157904	38.0	14.398524
3	4.0	3.804774	508.5	432.567932	1277.40	1149.375218	161.041667	68.731125	43.0	17.242457
4	4.0	3.932721	1111.0	308.353676	429.30	582.766886	61.166667	29.387183	48.0	17.558077

Figure 8. Boxplots & table of k-means clustering of each variable when k=5

Cluster 0: Occasional Customers

- Moderate engagement and recency
- Moderate CLV
- Average AUC
- Oldest cohort average with no customers younger than 44

Average except for age; targeting these consumers should be tailored to older audiences.

Cluster 1: Younger Regulars

- Purchases moderately often and relatively recently
- CLV in the mid-range
- Average spending habits with few outliers
- Significantly younger than average

A younger, engaged customer base, with opportunity to deepen relationships over time rather than re-engage. Their moderate CLV is constrained by lower average unit spend, suggesting targeted upselling could increase revenues as this cohort ages.

Cluster 2: Core Customers

- Highest purchase frequency, with some outliers purchasing extremely often
- Very recent purchasing
- Highest CLV with several extremely high value outliers
- Moderate AUC; value driven through volume only
- Broad Age Distribution

Most important customers to retain and understand due to lucrative habits. By far the most engaged spenders; likely a good target to increase revenue

Cluster 3: High-Ticket Occasional Buyers

- Lowest purchase frequency and low recency
- CLV is moderate despite low engagement
- Stems from AUC
- Broad age distribution, but average skews older

Individuals who are willing to spend a lot on one expensive item rather than many smaller items. A unique type of targeting is needed for these high price spenders.

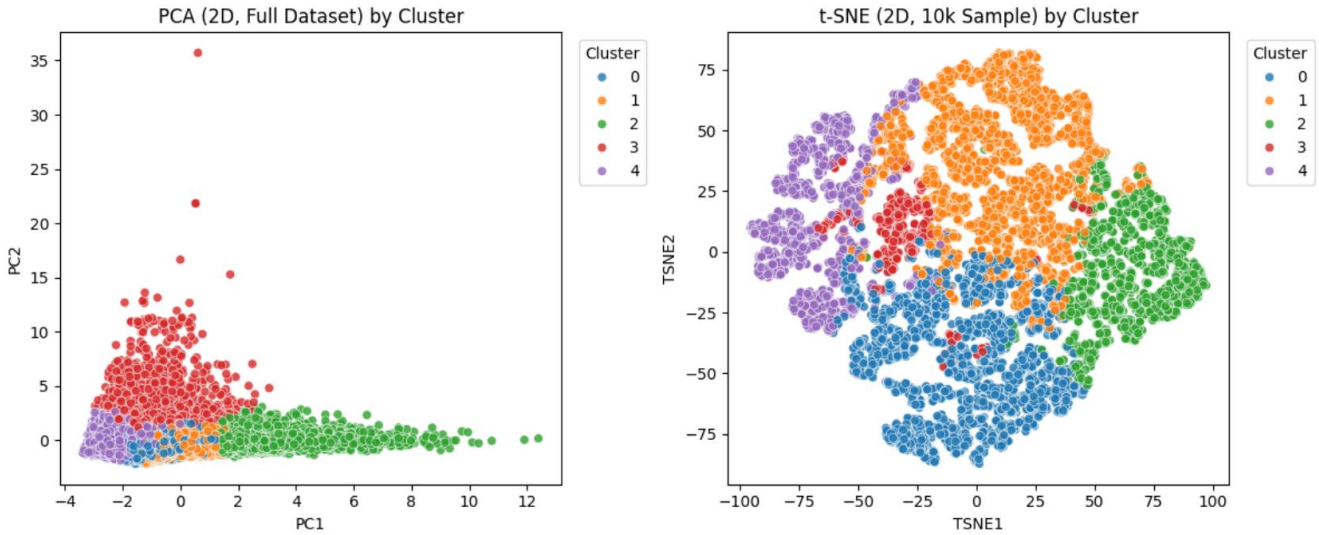
Cluster 4: Dormant Customers

- Low frequency and high recency
- Low CLV
- Average AUC
- Skews older but with wide range

Clear signs of disengagement, therefore, prime targets for re-engagement or de-prioritisation.

PCA & t-SNE

Figure 9. PCA & t-SNE on k=5 clusters



PCA showed an explained variance 0.65; the two components make up a significant proportion of the variance in the data. Graphically, there is overlap region for each cluster, especially for clusters 0, 1 & 4. Cluster 2 has some overlap but covers a large portion of the width on PC1, cluster 3 large range of height on PC2. This indicates moderate clusterisation effectiveness.

t-SNE clusters appear well grouped, with significant overlap mainly only for cluster 3. Each other component occupies distinct regions. This indicates the clustering captures local structure well, meaning we can be reasonably confident that customers within clusters share similar shopping habits.

Conclusions

Five clusters is an appropriate number to show meaningful differences in average shopper habits, and we are able to, from this, take five distinct customer profiles to present back to the business.

There are some limitations, for example, I was only able to sample 10,000 of the 68,000 customers. Additionally, other variables weren't included in the analysis, which may mean some part of the picture is missing. For example, the variable AUC doesn't take into consideration average basket size, a similar but distinctive metric.

Every customer can now be classified with a profile through their customer ID; the actionable insights of each cohort defined above can support the retailer to tailor their targeting strategy to each customer. The business should also validate whether these match any of their previous customer groupings and then work with industry experts to turn these insights into a strategy.

References

SAS, 2024. CUSTOMERS_CLEAN [Data set]. SAS. Last revised on 15 December 2021. [Accessed 20 February 2024].

Shutterstock (2024) Customer segment analysis marketing advertising concept, target audience, consumer or user group, businessman analyzing customer segment pie chart [Online image]. Shutterstock. (Accessed: 18 January 2026).